

Lab 08 Information:

- The material in this section is informational. Please read through the section as it helps you work on the lab exercises in the next section. There may be code examples in this informational section. You are welcome to copy-and-paste them to MATLAB to run the code, but no submission is needed on any test run.

Plotting Frequency Response in MATLAB: One can easily plot the frequency response $H(e^{j\hat{\omega}})$ of any LTI system in MATLAB. The important thing to remember is that $H(e^{j\hat{\omega}})$ is a complex-valued function of $\hat{\omega}$; hence we need to use two plots to completely describe $H(e^{j\hat{\omega}})$. Typically, we plot the magnitude response $|H(e^{j\hat{\omega}})|$ and the phase response $\angle H(e^{j\hat{\omega}})$ versus $\hat{\omega}$. Note also that $H(e^{j\hat{\omega}})$ is periodic in $\hat{\omega}$ with period 2π ; thus we only need to plot one period of $H(e^{j\hat{\omega}})$. Typically, we plot from $\hat{\omega} = -\pi$ to π .

For an example, consider the first difference filter specified by the difference equation:

$$y[n] = x[n] - x[n - 1].$$

We have worked out in the lecture videos that its frequency response is

$$H(e^{j\hat{\omega}}) = 1 - e^{-j\hat{\omega}}.$$

To plot the magnitude and phase response, we need to first create a frequency vector:

```
w = -pi:pi/1000:pi;
```

The increment in w above specifies how fine (so how fine?) we evaluate $H(e^{j\hat{\omega}})$ between $\hat{\omega} = -\pi$ to π . Then we can plot the responses using the following piece of code:

```
H = 1-exp(-1j*w); % FREQUENCY RESPONSE

figure;
subplot(2, 1, 1)
plot(w/pi, abs(H));
grid on;
title('Magnitude Response')
xlabel('Normalized Radian Frequency (\times \pi rad/sample)');
ylabel('Amplitude');
subplot(2, 1, 2)
plot(w/pi, angle(H)/pi);
grid on;
title('Phase Response')
xlabel('Normalized Radian Frequency (\times \pi rad/sample)');
ylabel('Phase (\times \pi rad)');
```

It is sometimes more desirable to plot the magnitude response in dB scale. To do so, we can replace the corresponding lines of code above by

```
plot(w/pi, 20*log10(abs(H)));  
ylabel('Amplitude (dB)');
```

The MATLAB function `freqz` also calculates and plots the frequency response of an FIR filter (can do actually beyond FIR filters). You may use it to check your frequency response calculation and plots. Use the `help` command to learn more about the `freqz` function.

Lab 08 Exercises:

- Unless stated otherwise, you must submit your solutions to all the lab exercises in this section.
- Your laboratory solutions should be submitted on Canvas as a single PDF. Use the skeleton code provide with the lab to answer each question.